

Electricity and Magnetism TEST

KU

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TI

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COM

10

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1. What is the net electrical force on the $2 \times 10^{-6} \text{ C}$ charge at the bottom? Although not labeled, assume the $-8 \times 10^{-6} \text{ C}$ is 5 cm away from the $2 \times 10^{-6} \text{ C}$. Round your final answer to 3 SD. (5 marks)

$$F_{AC} = \frac{kq_A q_C}{r^2}$$

$$= \frac{(9 \times 10^9 \text{ Nm}^2/\text{C}^2)(2 \times 10^{-6} \text{ C})(8 \times 10^{-6} \text{ C})}{(0.05 \text{ m})^2}$$

$$= 57.6 \text{ N } [\uparrow] \text{ (attract)}$$

$$F_{AB} = \frac{(9 \times 10^9 \text{ Nm}^2/\text{C}^2)(2 \times 10^{-6} \text{ C})(4 \times 10^{-6} \text{ C})}{(0.05 \text{ m})^2}$$

$$= 28.8 \text{ N } [\rightarrow 45^\circ \downarrow] \text{ (repel)}$$

$$F_{AC} = 28.8 \text{ N } [\rightarrow 45^\circ \uparrow] \text{ (attract)}$$

} y comp cancel out

$$x_T = F_{AB_x} + F_{AC_x}$$

$$= 2(28.8 \text{ N})(\sin 45^\circ)$$

$$= 40.729 \text{ N } [\rightarrow]$$

$$y_T = F_{AC}$$

$$= 57.6 \text{ N } [\uparrow]$$

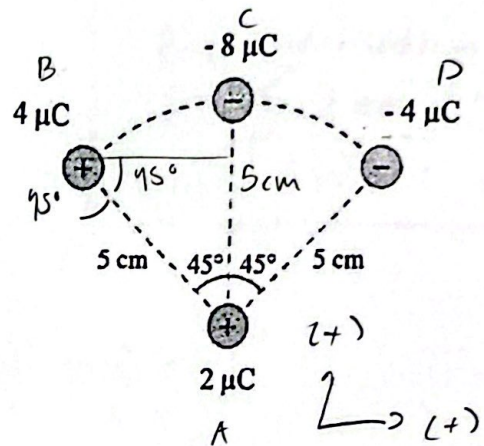
$$F_{NET} = \sqrt{40.729^2 + 57.6^2}$$

$$= 70.545 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{57.6}{40.729} \right)$$

$$= 54.7^\circ [\rightarrow 54.7^\circ \uparrow]$$

F_{NET} on the $2 \times 10^{-6} \text{ C}$ charge
is $70.5 \text{ N } [E 55^\circ N]$



5

2. An electron moving at a velocity of $6.7 \times 10^5 \text{ m/s}$ [E], horizontal to Earth's surface, enters a magnetic field with a magnitude of 3.2 T directed at an angle of 47° to the direction of motion upward in the vertical plane. What are the magnitude and direction of the magnetic force of the electron? Provide direction with respect to directions given in the questions. (3 marks)

$$v = 6.7 \times 10^5 \text{ m/s [E]}$$

$$B = 3.2 \text{ T}$$

$$\theta = 47^\circ$$

$$q = -1.60 \times 10^{-19} \text{ C}$$

$$F = qvB \sin \theta$$

$$F = (-1.60 \times 10^{-19} \text{ C})(6.7 \times 10^5 \text{ m/s})(3.2 \text{ T})(\sin 47^\circ)$$

$$= 2.5088 \times 10^{-13} \text{ N}$$

$$= 2.51 \times 10^{-13} \text{ N}$$

3/3

the magnitude
is $2.51 \times 10^{-13} \text{ N}$
the direction
is up AKA [N]
to the Earth's surface

3. The magnitude of the force exerted on a 0.55 m wire in an electric motor is 0.85 N . The 25 A current in a wire passes at a 90° angle to a magnetic field. Calculate the magnetic field. (2 marks)

$$F = ILB \sin \theta$$

$$B = \frac{F}{IL \sin \theta}$$

$$= 0.85 \text{ N}$$

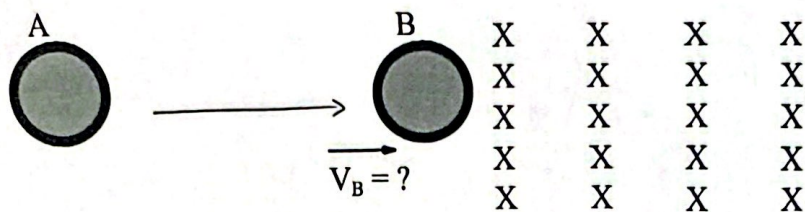
$$(0.55 \text{ m})(25 \text{ A})(\sin 90^\circ)$$

$$= 0.0618 \text{ T}$$

2/2

5

A particle has a mass of $1.80 \times 10^{-4} \text{ kg}$ and a charge of $+2.5 \times 10^{-5} \text{ C}$. It is released from rest at point A and accelerates until it reaches point B, moving on a horizontal straight line; note that the electric potential at A is 25 V greater than at point B. At point B, the charge then enters a magnetic field directed into the page.



What is the magnetic force and direction on the particle as it enters the magnetic field? (5 marks)

$$V_B - V_A = -25 \text{ V}$$

$$B = ?$$

$$m = 1.80 \times 10^{-4} \text{ kg}$$

$$q = 2.5 \times 10^{-5} \text{ C}$$

$$F = qvB \sin \theta$$

$$(V_B - V_A) q = \Delta E_{PE} = -KE$$

$$(-25 \text{ V})(2.5 \times 10^{-5} \text{ C}) = \Delta E_{PE}$$

$$\Delta E_{PE} = -6.25 \times 10^{-4} \text{ J}$$

$$-\Delta E_{PE} = KE = \frac{1}{2} mv^2$$

$$\sqrt{\frac{-\Delta E_{PE}}{0.5 m}} = v$$

$$v = \sqrt{\frac{-(-6.25 \times 10^{-4} \text{ J})}{0.5 (1.8 \times 10^{-4} \text{ kg})}}$$

$$v = 2.6352 \text{ m/s} \rightarrow$$

$$E = -\frac{\Delta V}{\Delta s}$$

$$r = \frac{mv}{qB}$$

$$F = qvB \sin \theta$$

$$B = 0.0610 \text{ T}$$

NOT

$$F = (2.5 \times 10^{-5} \text{ C})(2.6352 \text{ m/s})(\sin 90^\circ)$$

$$= 6.588 \times 10^{-5} \text{ N}$$



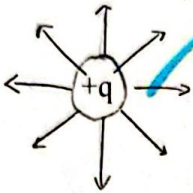
the direction is
due to up, the force
is $6.588 \times 10^{-5} \text{ N}$

10/10

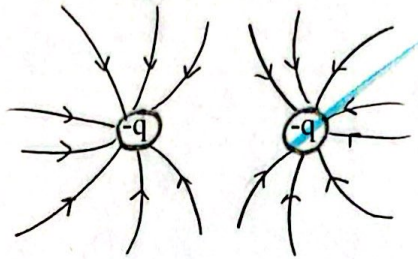
COMMUNICATION – 10 marks

1. Draw the electric field configuration around: (3 marks)

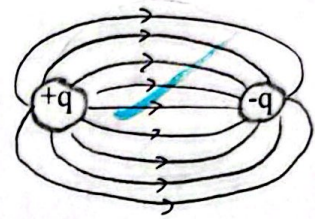
a) A small positive charge (1 mark)



b) Two negative charges (1 marks)

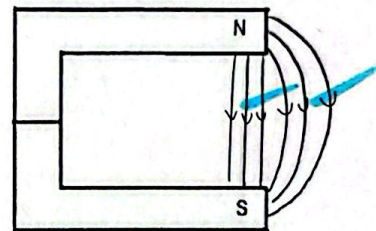


c) An electric dipole (1 mark)

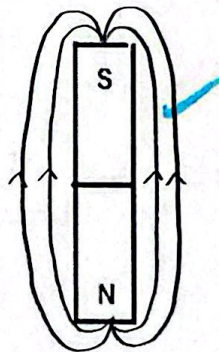
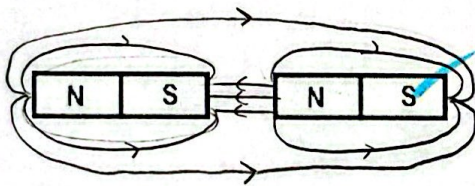


2. Draw the magnetic field lines of a horseshoe magnet. Clearly show in which area is the magnetic field lines the strongest with a couple of additional arrows. (2 marks)

3

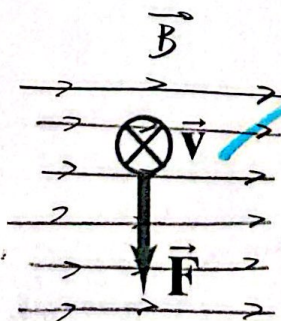


3. Draw the magnetic field lines for the following magnets. (3 marks)

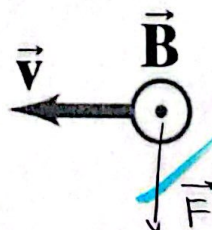


4. Add to the following diagrams what is required:

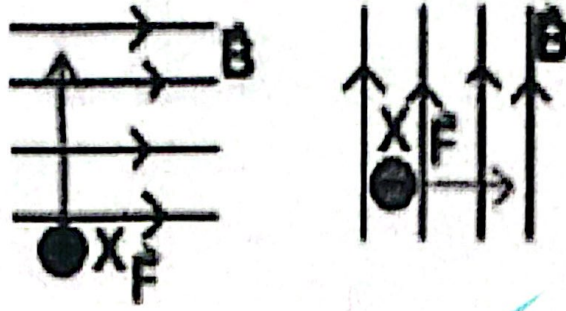
a) For a positive charge, the direction of $\vec{B}$ (1 mark)



b) For a negative charge, the direction of $\vec{F}$ (1 mark)



Q.2 Magnetic Force on Moving Charges



B- Electric Field: right

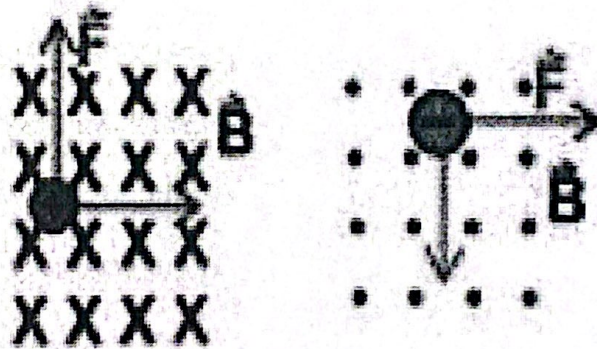
I- Current: up

F- Force: into

B: up

I: left

F: into



B: into

I: right

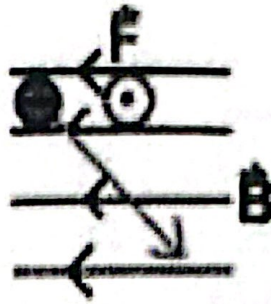
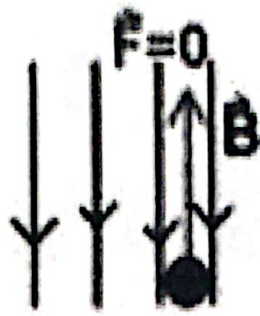
F: up

B: out of

I: up

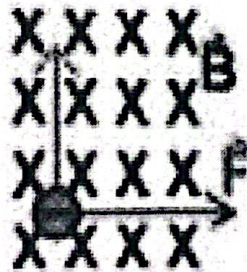
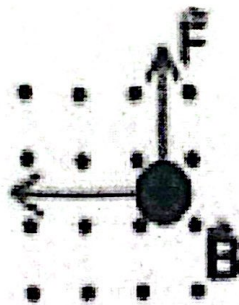
F: right

5/5



B: down
I: up
F: 0

B: left
I: up + left
F: out of



B: out of
I: left
F: up

B: into
I: down
F: right